| Method | Pretraining | Classification (macro-F1) | Detection (mAP) |
|---|---|---|---|
| Chance level | – | 16.67 | 7.84 |
| *Classical acoustic features* | | | |
| Spectral features | – | $33.52 \pm 2.06$ | $26.34 \pm 1.17$ |
| MFCCs | – | $44.20 \pm 2.20$ | $33.91 \pm 1.77$ |
| Mean spectrogram | – | $54.95 \pm 2.15$ | $47.80 \pm 2.10$ |
| *Off-the-shelf pretrained models* | | | |
| AVES-core | General audio (AudioSet, FSD50K) | $67.60 \pm 2.11$ | $57.39 \pm 2.07$ |
| BioLingual | Audio-text (AnimalSpeak) | $70.98 \pm 2.12$ | $66.52 \pm 2.23$ |
| AVES-bio | Animal vocalizations (AudioSet, VGGSound) | $75.13 \pm 1.96$ | $65.01 \pm 2.29$ |
| *Pretrained on OpenWhistle (ours)* | | | |
| AVES-bio$^{\ddagger}$ | OpenWhistle (from scratch) | $79.89 \pm 1.81$ | $76.17 \pm 1.97$ |
| wav2vec 2.0 | OpenWhistle (from scratch) | $\mathbf{81.40 \pm 1.78}$ | $\mathbf{79.59 \pm 1.83}$ |
| *Continual pretraining on OpenWhistle (ours)* | | | |
| AVES-bio$^{\dagger}$ | AVES-bio checkpoint $\rightarrow$ OpenWhistle | $75.24 \pm 2.01$ | $73.87 \pm 2.06$ |

Table 1: Performance on whistle-type classification (balanced, macro-F1) and whistle-type detection (mAP). Models are grouped by representation and pretraining strategy. All use linear probing with a logistic regression classifier on frozen representations. Scores are reported as percentages. Each entry gives the full-test metric ± the standard deviation over 1,000 bootstrap resamples of the test set. $^{\dagger}$Continual pretraining at 16 kHz with multi-band encoding; $^{\ddagger}$trained natively at 44.1 kHz.